

Assignment-10

Exercise 1: STAR Alignment

- Run STAR alignment for the remaining 4 samples using the provided reference genome and annotation file.
- Save the output as coordinate-sorted BAM files.

STAR (Spliced Transcripts Alignment to a Reference) is a fast and widely used RNA-seq read aligner. It is designed to map millions of short or long sequencing reads to a reference genome with high accuracy, particularly for eukaryotic organisms where spliced alignments are important. Unlike basic aligners, STAR can efficiently detect exon–exon junctions, making it suitable for transcriptome studies such as gene expression quantification, novel transcript discovery, and alternative splicing analysis.

cd 6_STAR (If not present, then mkdir 6_STAR)

mkdir 1_Index

mkdir 2_Alignment

Reference Indexing

cd 1_Index

STAR --runMode genomeGenerate --genomeDir . --genomeFastaFiles
Pfalciiparum.genome.fasta --sjdbGTFfile Pfalciiparum.genome.gtf

The above command tells STAR to run in **genomeGenerate** mode, which builds a genome index instead of aligning reads. The option **--genomeDir .** specifies the directory where the generated genome index will be stored (here, the current working directory). The **--genomeFastaFiles Pfalciiparum.genome.fasta** parameter provides the reference genome sequence in FASTA format, which is the base sequence STAR will index. The **--sjdbGTFfile Pfalciiparum.genome.gtf** option supplies the gene annotation in GTF format, so STAR can incorporate known splice junctions into the index, improving accuracy for RNA-seq spliced alignment. Together, these parameters prepare STAR to align sequencing reads efficiently in later steps.

```

ibab@LAPTOP-BVSTVK8Q:~/NGS/23aug/6_STAR/1_Index$ STAR --runMode genomeGenerate --genomeDir . --genomeFastaFiles /home/ibab/NGS/23aug/2_Reference/Pfalciparum.genome.fasta --sjdbGTFfile /home/ibab/NGS/23aug/2_Reference/Pfalciparum.genome.gtf /usr/lib/rna-star/bin/STAR-avx2 --runMode genomeGenerate --genomeDir . --genomeFastaFiles /home/ibab/NGS/23aug/2_Reference/Pfalciparum.genome.fasta --sjdbGTFfile /home/ibab/NGS/23aug/2_Reference/Pfalciparum.genome.gtf
STAR version: 2.7.10a compiled: 2022-01-16T16:35:44+00:00 <place not set in Debian package>
Sep 14 19:52:20 ..... started STAR run
Sep 14 19:52:20 ... starting to generate Genome files
Sep 14 19:52:23 ..... processing annotations GTF
!!!!!! WARNING: --genomeSAindexNbases 14 is too large for the genome size=23332839, which may cause seg-fault at the mapping step. Re-run genome generation with recommended --genomeSAindexNbases 11
Sep 14 19:52:24 ... starting to sort Suffix Array. This may take a long time...
Sep 14 19:52:24 ... sorting Suffix Array chunks and saving them to disk...
Sep 14 19:52:40 ... loading chunks from disk, packing SA...
Sep 14 19:52:40 ... finished generating suffix array
Sep 14 19:52:40 ... generating Suffix Array index
Sep 14 19:52:51 ... completed Suffix Array index
Sep 14 19:52:51 ..... inserting junctions into the genome indices
Sep 14 19:53:06 ... writing Genome to disk ...
Sep 14 19:53:06 ... writing Suffix Array to disk ...
Sep 14 19:53:07 ... writing SAindex to disk
Sep 14 19:53:22 ..... finished successfully

```

Indexing the reference genome using STAR creates below 16 different files including splice junction data.

```

ibab@LAPTOP-BVSTVK8Q:~/NGS/23aug/6_STAR/1_Index$ ls -lh
total 1.7G
-rw-r--r-- 1 ibab ibab 27M Sep 14 19:53 Genome
-rw-r--r-- 1 ibab ibab 33K Sep 14 19:53 Log.out
-rw-r--r-- 1 ibab ibab 198M Sep 14 19:53 SA
-rw-r--r-- 1 ibab ibab 1.5G Sep 14 19:53 SAindex
-rw-r--r-- 1 ibab ibab 121 Sep 14 19:52 chrLength.txt
-rw-r--r-- 1 ibab ibab 191 Sep 14 19:52 chrName.txt
-rw-r--r-- 1 ibab ibab 312 Sep 14 19:52 chrNameLength.txt
-rw-r--r-- 1 ibab ibab 138 Sep 14 19:52 chrStart.txt
-rw-r--r-- 1 ibab ibab 416K Sep 14 19:52 exonGeneInfo.tab
-rw-r--r-- 1 ibab ibab 164K Sep 14 19:52 exonInfo.tab
-rw-r--r-- 1 ibab ibab 245K Sep 14 19:52 geneInfo.tab
-rw-r--r-- 1 ibab ibab 760 Sep 14 19:53 genomeParameters.txt
-rw-r--r-- 1 ibab ibab 215K Sep 14 19:52 sjdbInfo.txt
-rw-r--r-- 1 ibab ibab 288K Sep 14 19:52 sjdbList.fromGTF.out.tab
-rw-r--r-- 1 ibab ibab 245K Sep 14 19:52 sjdbList.out.tab
-rw-r--r-- 1 ibab ibab 333K Sep 14 19:52 transcriptInfo.tab

```

Mapping

cd 2_Alignment

STAR --runMode alignReads --genomeDir /home/ibab/NGS/23aug/6_STAR/1_Index --readFilesIn /home/ibab/NGS/23aug/4_TrimGalore/SRR15276765/SRR15276765_1_val_1.fq.gz /home/ibab/NGS/23aug/4_TrimGalore/SRR15276765/SRR15276765_2_val_2.fq.gz --readFilesCommand gunzip -c --outFileNamePrefix SRR15276765 --runThreadN 2 --outSAMtype BAM SortedByCoordinate

This runs the aligner in **alignReads** mode, meaning it takes sequencing reads and maps them to reference genome. The parameter **--genomeDir /home/ibab/NGS/23aug/6_STAR/1_Index** points to the genome index built earlier. The option **--readFilesIn** provides the paired-end trimmed FASTQ files for the sample SRR15276765, while **--readFilesCommand gunzip -c** tells STAR to first decompress the .gz files before alignment. The **--outFileNamePrefix SRR15276765** ensures that all outputs are saved with this sample name. The **--runThreadN 2** allows STAR to use two CPU threads, improving speed. Finally, **--outSAMtype BAM SortedByCoordinate** outputs the alignment in BAM format, already sorted by genomic coordinates, which is convenient for downstream analyses such as visualization or counting.

Repeat the STAR alignment step for the remaining samples - SRR15276766, SRR15276767, SRR15276773.

```

ibab@LAPTOP-BVSTVK8Q:~/NGS/23aug/6_STAR/2_Alignment$ STAR --runMode alignReads --genomeDir /home/ibab/NGS/23aug/6_STAR/1_Index --readFilesIn /home/ibab/NGS/23aug/4_TrimGalore/SRR15276766/SRR15276766_1_val_1.fq.gz /home/ibab/NGS/23aug/4_TrimGalore/SRR15276766/SRR15276766_2_val_2.fq.gz --readFilesCommand gunzip -c --outFileNamePrefix SRR15276766 --runThreadN 2 --outSAMtype BAM SortedByCoordinate
/usr/lib/rna-star/bin/STAR-avx2 --runMode alignReads --genomeDir /home/ibab/NGS/23aug/6_STAR/1_Index --readFilesIn /home/ibab/NGS/23aug/4_TrimGalore/SRR15276766/SRR15276766_1_val_1.fq.gz /home/ibab/NGS/23aug/4_TrimGalore/SRR15276766/SRR15276766_2_val_2.fq.gz --readFilesCommand gunzip -c --outFileNamePrefix SRR15276766 --runThreadN 2 --outSAMtype BAM SortedByCoordinate
STAR version: 2.7.10a   compiled: 2022-01-16T16:35:44+00:00 <place not set in Debian package>
Sep 14 21:01:56 ..... started STAR run
Sep 14 21:01:56 ..... loading genome
Sep 14 21:02:33 ..... started mapping
Sep 15 03:45:02 ..... finished mapping
Sep 15 03:45:03 ..... started sorting BAM
Sep 15 04:30:39 ..... finished successfully

```

It generates five files; out of them, most important is `Aligned.sortedByCoord.out.bam` and `SJ.out.tab` files.

```
total 1.7G
-rw-r--r-- 1 ibab ibab 260M Sep 14 20:50 SRR15276765Aligned.sortedByCoord.out.bam
-rw-r--r-- 1 ibab ibab 2.0K Sep 14 20:50 SRR15276765Log.final.out
-rw-r--r-- 1 ibab ibab 7.1K Sep 14 20:50 SRR15276765Log.out
-rw-r--r-- 1 ibab ibab 2.9K Sep 14 20:50 SRR15276765Log.progress.out
-rw-r--r-- 1 ibab ibab 431K Sep 14 20:49 SRR15276765SJ.out.tab
-rw-r--r-- 1 ibab ibab 1.1G Sep 15 04:30 SRR15276766Aligned.sortedByCoord.out.bam
-rw-r--r-- 1 ibab ibab 2.0K Sep 15 04:30 SRR15276766Log.final.out
-rw-r--r-- 1 ibab ibab 7.1K Sep 15 04:30 SRR15276766Log.out
-rw-r--r-- 1 ibab ibab 4.3K Sep 15 04:30 SRR15276766Log.progress.out
-rw-r--r-- 1 ibab ibab 777K Sep 15 03:45 SRR15276766SJ.out.tab
```

samtools view SRR15276765Aligned.sortedByCoord.out.bam

[illegible]

SRR15276765SJ.out.tab file output -

Pf3D7_14_v3	3061578	3062261	2	2	0	3	0	54
Pf3D7_14_v3	3061595	3061743	1	1	1	38	0	54
Pf3D7_14_v3	3061895	3062072	1	1	1	24	0	61
Pf3D7_14_v3	3062462	3062559	1	1	1	26	0	54
Pf3D7_14_v3	3063273	3063374	2	2	1	25	0	56
Pf3D7_14_v3	3063471	3063679	2	2	1	34	0	61
Pf3D7_14_v3	3067611	3067722	1	1	1	62	0	63
Pf3D7_14_v3	3067806	3068059	1	1	1	101	0	60
Pf3D7_14_v3	3112690	3112754	0	0	0	3	0	55
Pf3D7_14_v3	3124988	3125026	1	1	0	3	0	12
Pf3D7_14_v3	3127521	3127745	1	1	0	1	0	37
Pf3D7_14_v3	3127804	3128031	1	1	0	5	0	55

- 1. Chromosome** – Reference sequence name (e.g., Pf3D7_14_v3) where the junction is located.
- 2. Intron Start** – First base of the intron (1-based coordinate).
- 3. Intron End** – Last base of the intron (1-based coordinate).

4. Strand – Strand of the splice junction:

- 0 = unstranded
- 1 = plus strand
- 2 = minus strand

5. Intron Motif – Canonical splice site motif:

- 0 = non-canonical
- 1 = GT/AG
- 2 = CT/AC
- 3 = GC/AG
- 4 = CT/GC
- 5 = AT/AC
- 6 = GT/AT

6. Annotated – Whether this junction is present in the supplied annotation:

- 0 = not annotated
- 1 = annotated

7. Unique Reads – Number of uniquely mapped reads crossing the junction.

8. Multi-mapped Reads – Number of multi-mapped reads crossing the junction.

9. Maximum Overhang – The length of the longest overhang (minimum number of bases on either side of the junction supporting it).

Quality check of STAR alignment using Qualimap

`qualimap bamqc -bam SRR15276765Aligned.sortedByCoord.out.bam -outfile SRR15276765.pdf -outformat pdf`

The above command evaluates the quality of RNA-seq alignment stored in a BAM file. The **bamqc** module is the specific tool within Qualimap that produces a comprehensive quality-control report of mapping results. The option **-bam SRR15276765Aligned.sortedByCoord.out.bam** tells Qualimap which BAM file to analyze—in this case, the aligned reads generated by STAR and sorted by genomic coordinates. Then, **-outfile SRR15276765.pdf** specifies the name of the final report you want to generate, so the output will be saved as a PDF file with that name instead of the default HTML format. Finally, **-outformat pdf** ensures the output is indeed produced in PDF format; without this option, the report would be in HTML by default.

```

ibab@LAPTOP-BVSTVK8Q:~/NGS/23aug/6_STAR/2_Alignment$ /home/ibab/NGS/Packages/qualimap_v2.3/qualimap_v2.3/qualimap bamqc \
-bam SRR15276765Aligned.sortedByCoord.out.bam \
-outfile SRR15276765.pdf \
-outformat pdf
Java memory size is set to 1200M
Launching application...

QualiMap v.2.3
Built on 2023-05-19 16:57

Selected tool: bamqc
Available memory (Mb): 35
Max memory (Mb): 998
Starting bam qc....
Loading sam header...
Loading locator...
Loading reference...
Number of windows: 400, effective number of windows: 415
Chunk of reads size: 1000
Number of threads: 4
Processed 50 out of 415 windows...
Processed 100 out of 415 windows...
Processed 150 out of 415 windows...
Processed 200 out of 415 windows...
Processed 250 out of 415 windows...

```

Repeat the qualimap step for the remaining samples - SRR15276766, SRR15276767, SRR15276773.

```

ibab@LAPTOP-BVSTVK8Q:~/NGS/23aug/6_STAR/2_Alignment/SRR15276765Aligned.sortedByCoord.out_stats$ ls -lh
total 928K
-rw-r--r-- 1 ibab ibab 917K Sep 15 05:48 SRR15276765.pdf
-rw-r--r-- 1 ibab ibab 6.0K Sep 15 05:48 genome_results.txt

```

SRR15276765

2. Summary

2.1. Globals

Reference size	23,332,839
Number of reads	4,303,239
Mapped reads	4,303,239 / 100%
Unmapped reads	0 / 0%
Mapped paired reads	4,303,239 / 100%
Mapped reads, first in pair	2,150,995 / 49.99%
Mapped reads, second in pair	2,152,244 / 50.01%
Mapped reads, both in pair	4,298,054 / 99.88%
Mapped reads, singletons	5,185 / 0.12%
Secondary alignments	411,808
Read min/max/mean length	20 / 126 / 133.63
Duplicated reads (estimated)	2,957,770 / 68.73%
Duplication rate	54.28%
Clipped reads	492,749 / 11.45%

2.2. ACGT Content

Number/percentage of A's	186,448,604 / 35.55%
Number/percentage of C's	75,111,290 / 14.32%
Number/percentage of T's	187,450,714 / 35.74%
Number/percentage of G's	75,403,768 / 14.38%
Number/percentage of N's	3,671 / 0%
GC Percentage	28.7%

2.3. Coverage

Mean	36.4237
Standard Deviation	213.6076

2.4. Mapping Quality

Mean Mapping Quality	143.13
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2.5. Insert size

Mean	493.28
Standard Deviation	8,846.15
P25/Median/P75	166 / 200 / 262

2.6. Mismatches and indels

Insertions	113,027
Mapped reads with at least one insertion	2.6%
Deletions	58,452
Mapped reads with at least one deletion	1.35%
Homopolymer indels	49.63%

2.7. Chromosome stats

Name	Length	Mapped bases	Mean coverage	Standard deviation
PF3D7_01_v3	640851	14056547	21.9342	69.8201
PF3D7_02_v3	947102	17665107	18.6517	38.7691

SRR15276766

2. Summary

2.1. Globals

Reference size	23,332,839
Number of reads	18,939,460
Mapped reads	18,939,460 / 100%
Unmapped reads	0 / 0%
Mapped paired reads	18,939,460 / 100%
Mapped reads, first in pair	9,471,033 / 50.01%
Mapped reads, second in pair	9,468,427 / 49.99%
Mapped reads, both in pair	18,922,218 / 99.91%
Mapped reads, singletons	17,242 / 0.09%
Secondary alignments	859,184
Read min/max/mean length	20 / 125 / 127.49
Duplicated reads (estimated)	15,867,755 / 83.78%
Duplication rate	69.21%
Clipped reads	1,088,557 / 5.75%

2.2. ACGT Content

Number/percentage of A's	841,507,252 / 36.56%
Number/percentage of C's	309,249,099 / 13.44%
Number/percentage of T's	841,024,978 / 36.54%
Number/percentage of G's	309,565,814 / 13.45%
Number/percentage of N's	234,232 / 0.01%
GC Percentage	26.89%

2.3. Coverage

Mean	172.6517
Standard Deviation	820.6423

2.4. Mapping Quality

Mean Mapping Quality	127.39
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2.5. Insert size

Mean	582.54
Standard Deviation	10,922.34
P25/Median/P75	194 / 231 / 298

2.6. Mismatches and indels

Insertions	250,688
Mapped reads with at least one insertion	1.32%
Deletions	245,516
Mapped reads with at least one deletion	1.29%
Homopolymer indels	50.13%

2.7. Chromosome stats

Name	Length	Mapped bases	Mean coverage	Standard deviation
PF3D7_01_v3	640851	72457475	113.0645	304.9301
PF3D7_02_v3	947102	97412503	102.8532	234.3651

SRR15276767

2. Summary

2.1. Globals

Reference size	23,332,839
Number of reads	21,779,412
Mapped reads	21,779,412 / 100%
Unmapped reads	0 / 0%
Mapped paired reads	21,779,412 / 100%
Mapped reads, first in pair	10,891,338 / 50.01%
Mapped reads, second in pair	10,888,074 / 49.99%
Mapped reads, both in pair	21,758,148 / 99.9%
Mapped reads, singletons	21,264 / 0.1%
Secondary alignments	740,590
Read min/max/mean length	20 / 125 / 126.1
Duplicated reads (estimated)	17,660,614 / 81.09%
Duplication rate	59.24%
Clipped reads	1,205,690 / 5.54%

2.2. ACGT Content

Number/percentage of A's	968,450,303 / 36.58%
Number/percentage of C's	355,260,574 / 13.42%
Number/percentage of T's	968,153,464 / 36.57%
Number/percentage of G's	355,047,136 / 13.41%
Number/percentage of N's	263,684 / 0.01%
GC Percentage	26.83%

2.3. Coverage

Mean	198.6472
Standard Deviation	982.5353

2.4. Mapping Quality

Mean Mapping Quality	123.62
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2.5. Insert size

Mean	592.53
Standard Deviation	11,262.64
P25/Median/P75	194 / 230 / 294

2.6. Mismatches and indels

Insertions	275,180
Mapped reads with at least one insertion	1.26%
Deletions	274,404
Mapped reads with at least one deletion	1.25%
Homopolymer indels	49.82%

2.7. Chromosome stats

Name	Length	Mapped bases	Mean coverage	Standard deviation
PF3D7_01_v3	640851	90920469	141.8746	419.2631
PF3D7_02_v3	947102	109078012	115.1703	262.6838

SRR15276773

2. Summary

2.1. Globals

Reference size	23,332,839
Number of reads	21,609,060
Mapped reads	21,609,060 / 100%
Unmapped reads	0 / 0%
Mapped paired reads	21,609,060 / 100%
Mapped reads, first in pair	10,805,359 / 50%
Mapped reads, second in pair	10,803,701 / 50%
Mapped reads, both in pair	21,590,804 / 99.92%
Mapped reads, singletons	18,256 / 0.08%
Secondary alignments	639,856
Read min/max/mean length	20 / 125 / 125.92
Duplicated reads (estimated)	17,539,194 / 81.17%
Duplication rate	56.07%
Clipped reads	1,134,189 / 5.25%

2.2. ACGT Content

Number/percentage of A's	968,920,766 / 36.79%
Number/percentage of C's	347,500,126 / 13.19%
Number/percentage of T's	969,645,183 / 36.82%
Number/percentage of G's	347,475,489 / 13.19%
Number/percentage of N's	272,267 / 0.01%
GC Percentage	26.39%

2.3. Coverage

Mean	194.9005
Standard Deviation	931.372

2.4. Mapping Quality

Mean Mapping Quality	126.06
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2.5. Insert size

Mean	586.66
Standard Deviation	10,920.18
P25/Median/P75	201 / 239 / 315

2.6. Mismatches and indels

Insertions	263,540
Mapped reads with at least one insertion	1.21%
Deletions	264,209
Mapped reads with at least one deletion	1.22%
Homopolymer indels	50.24%

2.7. Chromosome stats

Name	Length	Mapped bases	Mean coverage	Standard deviation
Pf3D7_01_v3	640851	70896520	110.6287	284.0836
Pf3D7_02_v3	947102	122903890	129.7684	392.5326

Detailed qualimap PDF (containing plots such as coverage histograms, duplication rate histogram, mapped reads nucleotide content, mapped reads GC content distribution, homopolymer indels, mapping quality histogram, insert size histogram, etc.) has been attached with this report for further analysis.

Quality check of multiple STAR alignments using MultiQC

- “.” means the folder containing qualimap results.
- MultiQC is a tool that aggregates results from multiple quality control analyses, including BAM file QC reports, into a single interactive report. It helps compare and summarize alignment statistics across different samples in one place. To run multiQC first we have to run qualimap for all samples.

```
ibab@LAPTOP-BVSTVK8Q:~/NGS/23aug/6_STAR/2_Alignment$ multiqc .  
  
/// MultiQC | v1.12  
  
multiqc | MultiQC Version v1.31 now available!  
multiqc | Search path : /home/ibab/NGS/23aug/6_STAR/2_Alignment  
searching | 100% 29/29  
qualimap | Found 4 BamQC reports  
star | Found 4 reports  
multiqc | Compressing plot data  
multiqc | Report : multiqc_report.html  
multiqc | Data : multiqc_data  
multiqc | MultiQC complete
```

```
ibab@LAPTOP-BVSTVK8Q:~/NGS/23aug/6_STAR/2_Alignment$ ls -lh
total 3.6G
-rw-r--r-- 1 ibab ibab 260M Sep 14 20:50 SRR15276765Aligned.sortedByCoord.out.bam
drwxr-xr-x 2 ibab ibab 4.0K Sep 15 05:48 SRR15276765Aligned.sortedByCoord.out_stats
-rw-r--r-- 1 ibab ibab 2.0K Sep 14 20:50 SRR15276765Log.final.out
-rw-r--r-- 1 ibab ibab 7.1K Sep 14 20:50 SRR15276765Log.out
-rw-r--r-- 1 ibab ibab 2.9K Sep 14 20:50 SRR15276765Log.progress.out
-rw-r--r-- 1 ibab ibab 431K Sep 14 20:49 SRR15276765SJ.out.tab
-rw-r--r-- 1 ibab ibab 1.1G Sep 15 04:30 SRR15276766Aligned.sortedByCoord.out.bam
drwxr-xr-x 2 ibab ibab 4.0K Sep 15 06:04 SRR15276766Aligned.sortedByCoord.out_stats
-rw-r--r-- 1 ibab ibab 2.0K Sep 15 04:30 SRR15276766Log.final.out
-rw-r--r-- 1 ibab ibab 7.1K Sep 15 04:30 SRR15276766Log.out
-rw-r--r-- 1 ibab ibab 4.3K Sep 15 04:30 SRR15276766Log.progress.out
-rw-r--r-- 1 ibab ibab 777K Sep 15 03:45 SRR15276766SJ.out.tab
-rw-r--r-- 1 ibab ibab 1.2G Sep 15 05:21 SRR15276767Aligned.sortedByCoord.out.bam
drwxr-xr-x 2 ibab ibab 4.0K Sep 15 06:19 SRR15276767Aligned.sortedByCoord.out_stats
-rw-r--r-- 1 ibab ibab 2.0K Sep 15 05:21 SRR15276767Log.final.out
-rw-r--r-- 1 ibab ibab 7.1K Sep 15 05:21 SRR15276767Log.out
-rw-r--r-- 1 ibab ibab 4.3K Sep 15 05:21 SRR15276767Log.progress.out
-rw-r--r-- 1 ibab ibab 980K Sep 15 05:17 SRR15276767SJ.out.tab
-rw-r--r-- 1 ibab ibab 1.2G Sep 15 06:14 SRR15276773Aligned.sortedByCoord.out.bam
drwxr-xr-x 2 ibab ibab 4.0K Sep 15 06:29 SRR15276773Aligned.sortedByCoord.out_stats
-rw-r--r-- 1 ibab ibab 2.0K Sep 15 06:14 SRR15276773Log.final.out
-rw-r--r-- 1 ibab ibab 7.1K Sep 15 06:14 SRR15276773Log.out
-rw-r--r-- 1 ibab ibab 4.2K Sep 15 06:14 SRR15276773Log.progress.out
-rw-r--r-- 1 ibab ibab 956K Sep 15 06:05 SRR15276773SJ.out.tab
drwxr-xr-x 2 ibab ibab 4.0K Sep 15 06:42 multiqc_data
-rw-r--r-- 1 ibab ibab 1.2M Sep 15 06:42 multiqc_report.html
```



A modular tool to aggregate results from bioinformatics analyses across many samples into a single report.

Report generated on 2025-09-15, 06:42 based on data in: `/home/ibab/NGS/23aug/6_STAR/2_Alignment`

Welcome! Not sure where to start?

[Watch a tutorial video](#)

(6:06)

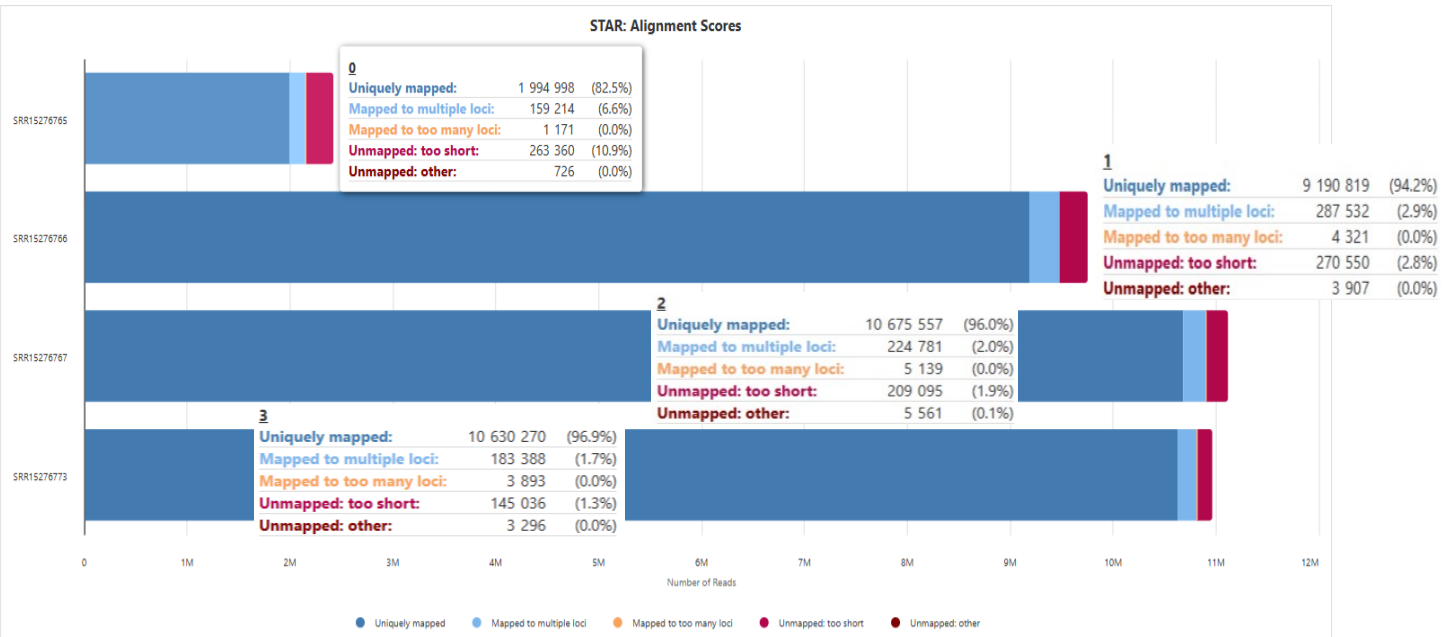
General Statistics

Copy table Configure Columns Plot Showing ⁴/₄ rows and ⁴/₇ columns.

Sample Name	Mean cov	% Aligned	% Aligned	M Aligned
SRR15276765	36.4X	100.0%	82.5%	2.0
SRR15276766	172.7X	100.0%	94.2%	9.2
SRR15276767	198.6X	100.0%	96.0%	10.7
SRR15276773	194.9X	100.0%	96.9%	10.6

Alignment Scores

[Number of Reads](#) [Percentages](#)




```
ibab@LAPTOP-BVSTVK8Q:~/NGS/23aug/6_STAR/2_Alignment/multiqc_data$ cat multiqc_general_stats.txt
Sample QualiMap_mqc-generalstats-qualimap-mean_coverage QualiMap_mqc-generalstats-qualimap-percentage_aligned QualiMap_mqc-generalstats-qualimap-mapped_reads QualiMap_mqc-generalstats-qualimap-total_reads QualiMap_mqc-generalstats-qualimap-general_error_rate STAR_mqc-generalstats-star-uniquely_mapped_percent STAR_mqc-generalstats-star-uniquely_mapped
SRR15276765 36.4237 100.0 4303239.0 4303239.0 0.0 82.46 1994998.0
SRR15276766 172.6517 100.0 18939460.0 18939460.0 0.0 94.2 9190819.0
SRR15276767 198.6472 100.0 21779412.0 21779412.0 0.0 96.0 10675557.0
SRR15276773 194.9005 100.0 21609060.0 21609060.0 0.0 96.94 10630270.0
```

```
ibab@LAPTOP-BVSTVK8Q:~/NGS/23aug/6_STAR/2_Alignment/multiqc_data$ cat multiqc_qualimap_bamqc_genome_results.txt
Sample bam_file total_reads mapped_reads mapped_bases sequenced_bases mean_insert_size median_insert_size mean_mapping_quality
general_error_rate mean_coverage percentage_aligned
SRR15276765 SRR15276765Aligned.sortedByCoord.out.bam 4303239.0 4303239.0 849868920.0 524418047.0 493.2761 200.0 143.1259
0.0 36.4237 100.0
SRR15276766 SRR15276766Aligned.sortedByCoord.out.bam 18939460.0 18939460.0 4028454011.0 2301581375.0 582.5376 231.0 127.3887
0.0 172.6517 100.0
SRR15276767 SRR15276767Aligned.sortedByCoord.out.bam 21779412.0 21779412.0 4635003015.0 2647175161.0 592.5303 230.0 123.6224
0.0 198.6472 100.0
SRR15276773 SRR15276773Aligned.sortedByCoord.out.bam 21609060.0 21609060.0 4547582823.0 2633813831.0 586.6637 239.0 126.0642
0.0 194.9005 100.0
```

```
ibab@LAPTOP-BVSTVK8Q:~/NGS/23aug/6_STAR/2_Alignment/multiqc_data$ cat multiqc_star.txt
Sample total_reads avg_input_read_length uniquely_mapped uniquely_mapped_percent avg_mapped_read_length num_splices num_annotated_splices num_unmapped_tooshort_percent
GTAG_splices num_GCAG_splices num_ATAC_splices num_noncanonical_splices mismatch_rate deletion_rate deletion_length insertion_rate unmapped_mismatches_percent
SRR15276765 2419469.0 243.0 1994998.0 82.46 243.72 580406.0 558802.0 576212.0 1788.0 225.0 2181.0 0.13 0.01
1.52 0.02 1.17 159214.0 6.58 1171.0 0.05 0.0 10.88 0.03 0 263360 726
SRR15276766 9757129.0 243.0 9190819.0 94.2 242.89 3158318.0 3019870.0 3130404.0 11739.0 1191.0 14984.0 0.16 0.02
1.58 0.01 1.21 287532.0 2.95 4321.0 0.04 0.0 2.77 0.04 0 270550 3907
SRR15276767 11120133.0 243.0 10675557.0 96.0 242.95 3478834.0 3311383.0 3445845.0 13934.0 1216.0 17839.0 0.16 0.02
1.59 0.01 1.21 224781.0 2.02 5139.0 0.05 0.0 1.88 0.05 0 209095 5561
SRR15276773 10965883.0 244.0 10630270.0 96.94 243.66 4161966.0 3982230.0 4125702.0 17756.0 1405.0 17103.0 0.15 0.02
1.56 0.01 1.2 183388.0 1.67 3893.0 0.04 0.0 1.32 0.03 0 145036 3296
```

Exercise 2: Read Count with HISAT2 Alignments

- Perform read count analysis on HISAT2-aligned BAM files using htseq-count.
- Perform read count analysis on HISAT2-aligned BAM files using FeatureCount.
- For FeatureCount data:
 - ❖ Generate Boxplot, PCA, and PCC plots with and without CPM filtering.
 - ❖ Generate PCA and PCC plots using rlog filtering.

cd 5_HiSAT2

mkdir 3_htseq

mkdir 4_feature_count

Read counting in features with htseq-count

cd 3_htseq

```
htseq-count /home/ibab/NGS/23aug/5_HiSAT2/2_Alignment/SRR15276765_sorted.bam
/home/ibab/NGS/23aug/2_Reference/Pfalciparum.genome.gtf --idattr gene_id -t exon -f bam >
SRR15276765_htseq.txt
```

The above command uses HTSeq to count how many sequencing reads from the alignment map to each gene. The first argument is the sorted input BAM file that contains RNA-seq reads aligned to the genome using HISAT2. The second argument is the gene annotation file in GTF format, which tells HTSeq where the exons, genes, and other features are located. The option **--idattr gene_id**

specifies that the counts should be summarized by the `gene_id` field in the GTF file, meaning all reads overlapping exons belonging to the same gene ID will be grouped together. The **-t exon parameter** instructs HTSeq to count reads only over features labelled as “exon” in the annotation, since exons represent the transcribed portions of genes. The **-f bam flag** tells the program that the input alignment is in BAM format rather than SAM. Finally, it saves the resulting table of read counts per gene into the file `SRR15276765_htseq.txt`.

```
(base) ibab@IBAB-MScBDB2-Comp007:~/NGS/23_aug/5_HiSat2/3_htseq$ htseq-count /home/ibab/NGS/23aug/5_HiSat2/2_Alignment/SRR15276765_sorted.bam /home/ibab/NGS/23aug/2_Reference/Pfalciparum.genome.gtf --idattr gene_id -t exon -f bam > SRR15276765_htseq.txt
35240 GFF lines processed.
Warning: Read SRR15276765.1651800 claims to have an aligned mate which could not be found in an adjacent line.
100000 alignment record pairs processed.
200000 alignment record pairs processed.
300000 alignment record pairs processed.
400000 alignment record pairs processed.
500000 alignment record pairs processed.
600000 alignment record pairs processed.
700000 alignment record pairs processed.
800000 alignment record pairs processed.
```

Repeat the `htseq-count` step for remaining samples - `SRR15276765`, `SRR15276767` and `SRR15276773`.

```
(base) ibab@IBAB-MScBDB2-Comp007:~/NGS/23aug/5_HiSat2/3_htseq$ cat SRR15276765_htseq.txt
PF3D7_0100100 7
PF3D7_0100200 9
PF3D7_0100300 0
PF3D7_0100400 1
PF3D7_0100500 3
PF3D7_0100600 2
PF3D7_0100700 0
PF3D7_0100800 0
PF3D7_0100900 0
```

Column 1: feature IDs (e.g., `gene_id` from the GTF file).

Column 2: number of reads that mapped to that feature.

Diagnostic rows (at the end of the `.txt` file)

- ◆ `__no_feature`: reads aligned but did not overlap any annotated gene/exon.

If this number is very high, it can mean:

The annotation file (GTF) does not match the reference genome used for alignment, or the data contains a lot of intergenic reads.

- ◆ `__ambiguous`: reads overlapped more than one gene, so `htseq-count` could not decide.
- ◆ `__too_low_aQual`: Reads with low alignment quality (mapping quality < 10 by default) – excluded.
- ◆ `__not_aligned`: Reads in the BAM file that were not aligned to the genome at all.
- ◆ `__alignment_not_unique`: reads mapped to multiple genome locations.

SRR15276765

```
__no_feature      3690045
__ambiguous       3060
__too_low_aQual   17093
__not_aligned     271543
__alignment_not_unique 304430
```

SRR15276766

```
__no_feature      16474837
__ambiguous       21917
__too_low_aQual   152428
__not_aligned     340888
__alignment_not_unique 592616
```

SRR15276767

```
__no_feature      19103846
__ambiguous       26598
__too_low_aQual   180281
__not_aligned     279685
__alignment_not_unique 499068
```

SRR15276773

```
__no_feature      19381725
__ambiguous       22731
__too_low_aQual   178189
__not_aligned     215076
__alignment_not_unique 447503
```

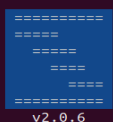
Read counting in features with featureCounts

cd 4_feature_count

```
featureCounts -p -t exon -g gene_id -a Pfalciparum.genome.gtf
/home/ibab/NGS/23aug/5_HiSat2/2_Alignment/SRR15276765_sorted.bam -o
SRR15276765.txt
```

This command runs **featureCounts**, a read summarization tool from the ‘Subread’ package, to count how many sequencing reads map to genomic features. The **option -p** tells it to treat reads as paired-end fragments instead of individual reads, so both mates of a pair are considered together when assigning to features. The **-t exon option** specifies that the program should count reads overlapping exon features (as defined in the annotation file). The **-g gene_id option** tells featureCounts to group exons by their gene_id attribute in the GTF file, meaning all exons belonging to the same gene are aggregated to produce one count per gene. The **-a flag** provides the annotation file (Pfalciparum.genome.gtf), which defines the coordinates and attributes of features like exons and genes. The next argument is the BAM file (SRR15276765_sorted.bam), which contains the aligned reads from HISAT2; featureCounts scans this file to assign reads to genes. Finally, the -o SRR15276765.txt option specifies the output file, where the resulting count table (with one row per gene and its associated read count) will be saved.

```
(base) ibab@IBAB-MScBDB2-Comp007:~/NGS/23aug/5_HiSat2/4_feature_count$ featureCounts -p -t exon -g gene_id -a /home/ibab/NGS/23aug/2_Reference/Pfalciparum.genome.gtf /home/ibab/NGS/23aug/5_HiSat2/2_Alignment/SRR15276765_sorted.bam -o SRR15276765.txt
```



SUBREAD

v2.0.6

```
//===== featureCounts setting =====\\
||
|| Input files : 1 BAM file
||               SRR15276765_sorted.bam
||
|| Output file : SRR15276765.txt
||               Summary : SRR15276765.txt.summary
||               Paired-end : yes
||               Count read pairs : no
||               Annotation : Pfalciparum.genome.gtf (GTF)
||               Dir for temp files : ./
||
|| Threads : 1
||               Level : meta-feature level
||               Multimaping reads : not counted
||               Multi-overlapping reads : not counted
||               Min overlapping bases : 1
||
```

```
(base) ibab@IBAB-MScBDB2-Comp007:~/NGS/23aug/5_HiSat2/4_feature_count$ ls -lh
total 2.2M
-rw-rw-r-- 1 ibab ibab 544K Sep 15 18:20 SRR15276765.txt
-rw-rw-r-- 1 ibab ibab 426 Sep 15 18:20 SRR15276765.txt.summary
-rw-rw-r-- 1 ibab ibab 547K Sep 15 18:27 SRR15276766.txt
-rw-rw-r-- 1 ibab ibab 429 Sep 15 18:27 SRR15276766.txt.summary
-rw-rw-r-- 1 ibab ibab 547K Sep 15 18:29 SRR15276767.txt
-rw-rw-r-- 1 ibab ibab 430 Sep 15 18:29 SRR15276767.txt.summary
-rw-rw-r-- 1 ibab ibab 547K Sep 15 18:30 SRR15276773.txt
-rw-rw-r-- 1 ibab ibab 429 Sep 15 18:30 SRR15276773.txt.summary
```

```
# Program:featureCounts v2.0.6; Command:"featureCounts" "-p" "-t" "exon" "-g" "gene_id" "-a" "/home/ibab/NGS/23aug/2_Reference/PfalciParum.genome.gtf"
Geneid Chr Start End Strand Length /home/ibab/NGS/23aug/5_HiSat2/2_Alignment/SRR15276765_sorted.bam
malmito_rna_16:rRNA M76611 3 33 + 31 0
malmito_rna_20:rRNA M76611 34 71 + 38 0
malmito_rna_9:rRNA M76611 72 125 + 54 0
malmito_rna_17:rRNA M76611 126 165 + 40 0
malmito_rna_LSUC:rRNA M76611 204 226 - 23 0
malmito_rna_LSUG:rRNA M76611 283 389 - 107 0
malmito_SSUB:rRNA M76611 390 505 - 116 0
malmito_rna_1:rRNA M76611 506 593 - 88 0
malmito_rna_15:rRNA M76611 594 624 - 31 0
malmito_rna_10:rRNA M76611 625 724 - 100 0
```

SRR15276765.txt.summary

SRR15276766.txt.summary

```
Status /home/ibab/NGS/23aug/5_HiSat2/2_Alignment/SRR15276765_sorted.bam
Assigned 3795038
Unassigned_Unmapped 549819
Unassigned_Read_Type 0
Unassigned_Singleton 0
Unassigned_MappingQuality 0
Unassigned_Chimera 0
Unassigned_FragmentLength 0
Unassigned_Duplicate 0
Unassigned_MultiMapping 692281
Unassigned_Secondary 0
Unassigned_NonSplit 0
Unassigned_NoFeatures 139582
Unassigned_Overlapping_Length 0
Unassigned_Ambiguity 49077
```

```
Status /home/ibab/NGS/23aug/5_HiSat2/2_Alignment/SRR15276766_sorted.bam
Assigned 17094799
Unassigned_Unmapped 715339
Unassigned_Read_Type 0
Unassigned_Singleton 0
Unassigned_MappingQuality 0
Unassigned_Chimera 0
Unassigned_FragmentLength 0
Unassigned_Duplicate 0
Unassigned_MultiMapping 1473250
Unassigned_Secondary 0
Unassigned_NonSplit 0
Unassigned_NoFeatures 853029
Unassigned_Overlapping_Length 0
Unassigned_Ambiguity 243217
```

SRR15276767.txt.summary

SRR15276773.txt.summary

```
Status /home/ibab/NGS/23aug/5_HiSat2/2_Alignment/SRR15276767_sorted.bam
Assigned 19823361
Unassigned_Unmapped 607797
Unassigned_Read_Type 0
Unassigned_Singleton 0
Unassigned_MappingQuality 0
Unassigned_Chimera 0
Unassigned_FragmentLength 0
Unassigned_Duplicate 0
Unassigned_MultiMapping 1323309
Unassigned_Secondary 0
Unassigned_NonSplit 0
Unassigned_NoFeatures 1008990
Unassigned_Overlapping_Length 0
Unassigned_Ambiguity 287221
```

```
Status /home/ibab/NGS/23aug/5_HiSat2/2_Alignment/SRR15276773_sorted.bam
Assigned 19806135
Unassigned_Unmapped 482666
Unassigned_Read_Type 0
Unassigned_Singleton 0
Unassigned_MappingQuality 0
Unassigned_Chimera 0
Unassigned_FragmentLength 0
Unassigned_Duplicate 0
Unassigned_MultiMapping 1200802
Unassigned_Secondary 0
Unassigned_NonSplit 0
Unassigned_NoFeatures 888507
Unassigned_Overlapping_Length 0
Unassigned_Ambiguity 302025
```

Visualisation of raw reads

Raw reads can be visualized using box plots, PCA, and PCC for quality assessment. Box plots of raw counts help identify differences in sequencing depth and overall data distribution among samples. PCA on raw data is important because the primary source of variance (PC1)

is likely driven by sequencing depth, allowing you to assess batch effects and technical biases before normalization. Finally, calculating the Pearson Correlation Coefficient on raw reads helps detect samples with poor correlation, which may indicate issues such as contamination or failed experiments early in the process.

For proceeding with normalization, we need raw_reads.txt as input file but we have .txt as output file that contains columns like gene ID, chromosome ID, start & end coordinates, length, read count. We want only the gene ID and read count in our raw_reads.txt file. So, we use following commands to extract the required column and then merge all the samples to get final file.

Command	Purpose
<pre>awk 'NR==1 {print \$1"\t"\$NF; next} /^PF3D7/ {print \$1"\t"\$NF}' SRR15276765.txt > PF3D7_sample1.txt</pre> Repeat for other samples also.	Extracts the Geneid (first column) and the raw read count (last column) from the featureCounts file. Keeps the header row (NR==1) and only rows where the gene ID starts with PF3D7. Saves into a new file.
<pre>paste PF3D7_sample1.txt PF3D7_sample2.txt PF3D7_sample3.txt PF3D7_sample4.txt > merged.txt</pre>	Combines all filtered files side by side into one file. At this stage, multiple Geneid columns are still present.
<pre>cut -f1,2,4,6,8 merged.txt > PF3D7_counts.txt</pre>	Retains only the first Geneid column and the count columns from each sample, producing a clean count matrix.
<pre>sed -i '1s/.*\/Geneid\tSRR15276765\ tSRR15276766\tSRR15276767\ tSRR15276773/' PF3D7_counts.txt</pre>	Replaces the messy header with a clear header line containing the sample names.
<pre>column -t PF3D7_counts.txt > raw_reads.txt</pre>	Properly aligned file (with proper tab spaces)

```
(base) ibab@IBAB-MScBDB2-Comp007:~/NGS/23aug/5_HiSat2/4_feature_count$ wc -l raw_reads.txt
5681 raw_reads.txt
(base) ibab@IBAB-MScBDB2-Comp007:~/NGS/23aug/5_HiSat2/4_feature_count$ head -5 raw_reads.txt
Geneid      SRR15276765  SRR15276766  SRR15276767  SRR15276773
PF3D7_0100100  138          957          952          791
PF3D7_0100200  55           222          75           122
PF3D7_0100300  4            6            6            2
PF3D7_0100400  12           65           107          122
```

Change the input file name to raw_reads.txt and update the no. of samples (GroupA - 2, GroupB - 2) also ([nano BOX_PCA_PCC_without_CPM_filtering](#)).

```
# Input: raw read count file
# Assumes first column is GeneID, rest are samples
#-----
counts <- read.table("raw_reads.txt", header=TRUE, row.names=1, sep="\t", check.names=FALSE)
```



```
# Define sample groups (adjust to match your samples!)
#-----
# Example: first 2 samples = GroupA, next 2 samples = GroupB
groups <- factor(c(rep("GroupA", 2), rep("GroupB", 2)))
names(groups) <- colnames(cpm_data)
```

Since, raw_reads.txt is tab or white-space separated file.

```
# Assumes first column is geneid, rest are samples
#-----
counts <- read.table("raw_reads.txt", header=TRUE, row.names=1, sep=" ", check.names=FALSE)
```

Rscript BOX_PCA_PCC_without_CPM_filtering.R

```
(base) ibab@IBAB-MScBDB2-Comp007:~/NGS/23aug/5_HiSat2/3_htseq$ Rscript BOX_PCA_PCC_without_CPM_filtering.R
Loading required package: limma
null device
1
✔ All plots saved in QC_plots.pdf
```

It generates a PDF file named Pre_QC_plots.pdf, containing all three plots along with a scree plot that explains the percentage of variance for each principal component.

Normalisation of raw read

Normalization is crucial for raw reads to account for technical differences in sequencing depth and library size between samples. This process makes gene counts comparable, ensuring that observed differences in gene expression reflect true biological variation rather than differences in total read counts.

To reduce noise from lowly expressed genes, CPM filtering (adjusts raw counts based on library size) is performed.

CPM filtering rule:

We require a gene to have $CPM > 1$ in at least N samples. N is usually set to the size of the smallest group, so that genes expressed in at least all replicates of one condition are retained. Here: GroupA has 2 samples, GroupB has 2 samples, so N can be set to 2 (lenient) or 3 (stricter).

nano BOX_PCC_PCA_with_CPM_filtering.R

```
#-----
# Step 2: CPM filtering
#-----
cpm_counts <- cpm(dge)
keep <- rowSums(cpm_counts > 1) >= 3
dge_filtered <- dge[keep, , keep.lib.sizes=FALSE]

cat("Genes before filtering:", nrow(dge), "\n")
cat("Genes after filtering:", nrow(dge_filtered), "\n")

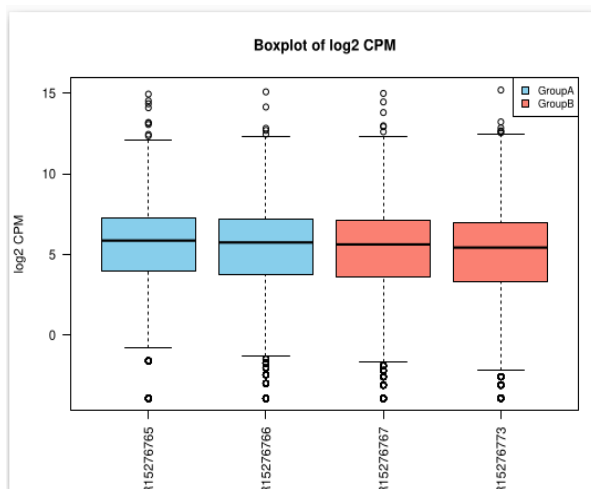
log_cpm <- cpm(dge_filtered, log=TRUE)
```

Rscript BOX_PCC_PCA_with_CPM_filtering.R

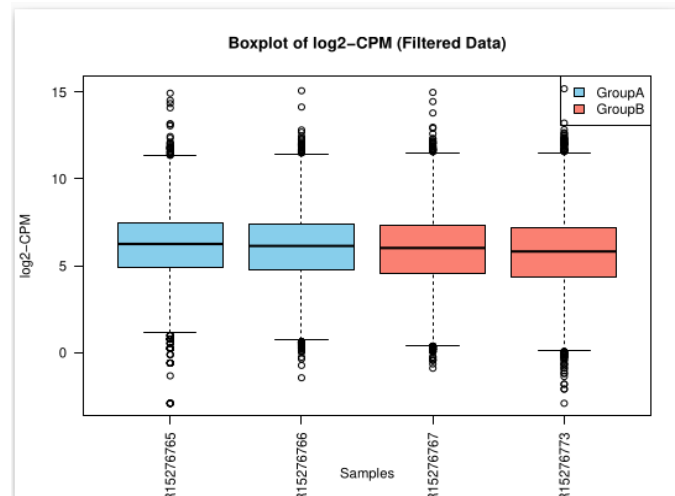
```
(base) ibab@IBAB-MScBDB2-Comp007:~/NGS/23aug/5_HiSat2/4_feature_count$ Rscript BOX_PCC_PCA_with_CPM_filtering.R
Loading required package: limma
Genes before filtering: 5680
Genes after filtering: 4937
null device
1
```

Comparison:

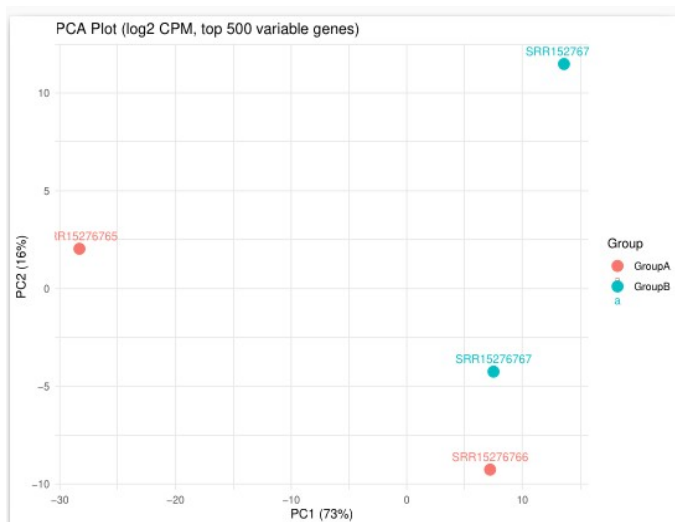
Without CPM Filtering



With CPM Filtering



The boxplot without CPM filtering shows the distribution of log2 CPM values for all genes, including many that are expressed at very low levels or not at all. This results in a wider spread of values at the lower end and more outliers near zero, reflecting background noise. After applying CPM filtering, lowly expressed genes are removed, leading to cleaner distributions. The median log2 CPM values shift slightly higher, and the overall spread becomes more comparable across samples. This indicates that filtering reduces noise and improves the consistency of expression profiles between samples, which is important for reliable downstream analyses such as PCA, clustering, and differential expression.

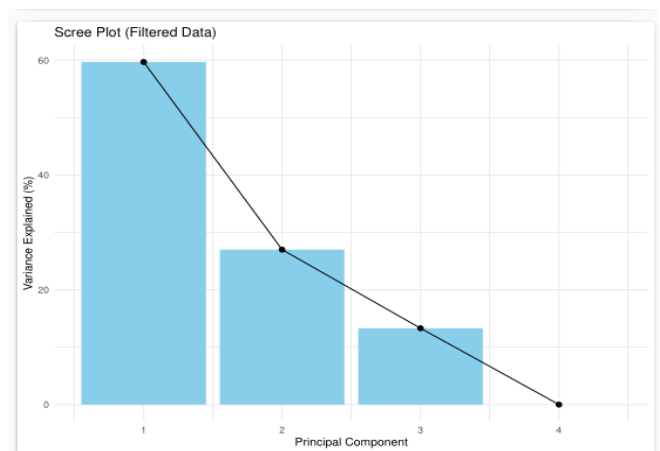
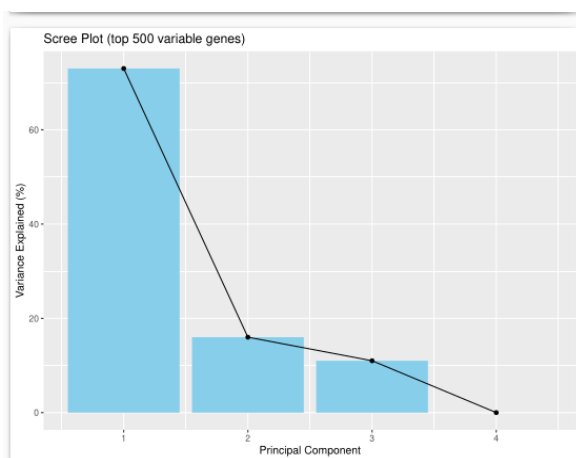


Without CPM filtering:

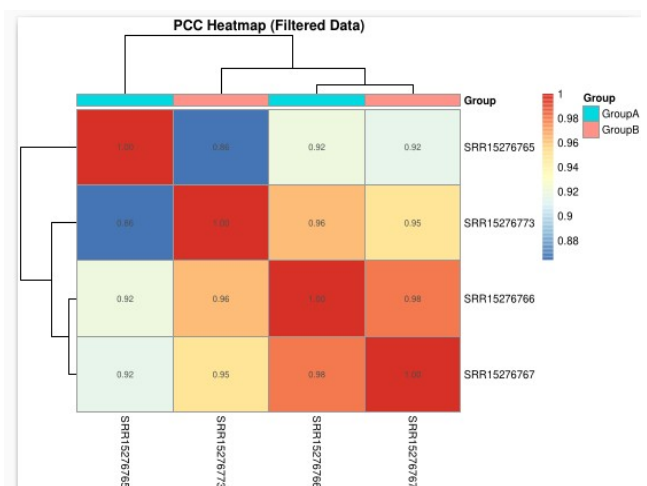
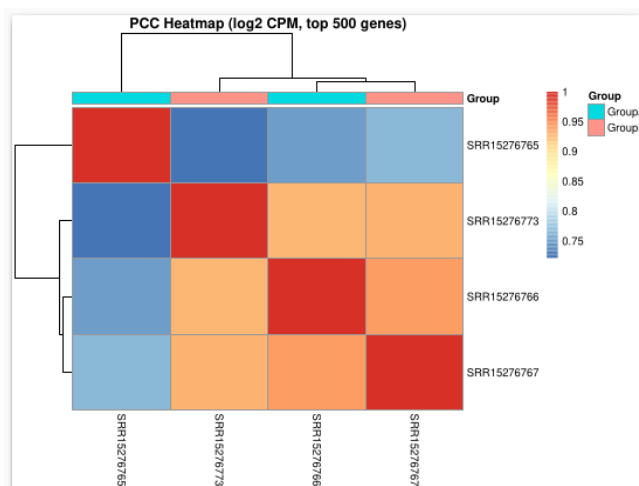
PC1 explains 73% of the variation, but the samples are spread apart in different directions. Even samples from the same group (GroupA: SRR15276765 & SRR15276766; GroupB: SRR15276767 & SRR15276773) don't sit close to each other. This happens because low-count/noisy genes dominate the variance.

With CPM filtering:

- PC1 explains 59.7% and PC2 27%, together capturing ~87% of the variation. Now, GroupA samples (SRR15276765 & SRR15276766) are on the left side, relatively closer to each other. GroupB samples (SRR15276767 & SRR15276773) are on the right side, again closer together. The clustering is not “overlapping” but more group-wise consistent compared to the without CPM filtering one.



In the scree plot without CPM filtering, the first principal component (PC1) explains a very large proportion of the variance (~73%), while the remaining components explain only small fractions. This suggests that the high variability in low-count or noisy genes is driving most of the signal, which may mask meaningful biological variation. After CPM filtering, the variance is distributed more evenly across the first few components, with PC1 explaining ~60% and PC2 explaining ~27%. This indicates that filtering reduces noise and allows additional components to capture relevant biological differences. Thus, CPM filtering leads to a more balanced variance distribution.



In the PCC heatmap without CPM filtering, the within-group correlations are relatively weak, with GroupA replicates showing values around 0.75–0.8 and GroupB replicates around 0.8–0.85. Cross-group correlations are of a similar magnitude, making it difficult to distinguish clear group-specific clustering. After CPM filtering, the correlation values improves: GroupA replicates correlate at ~0.92–0.95, and GroupB replicates show very strong correlations of ~0.96–0.98. At the same time, cross-group correlations drop to ~0.86–0.9, creating a sharper separation between groups. This demonstrates that CPM filtering reduces the noise contributed by lowly expressed genes, strengthens within-group consistency, and improves the biological interpretability of the data.

rlog (Regularized Log Transformation):

- Purpose: rlog is designed to stabilize variance across samples, especially when count data span a wide range (low counts vs very high counts).
- Comparison to log2(CPM): Unlike a simple log2 transformation, rlog shrinks values for lowly expressed genes more strongly, reducing the dominance of noise.
- Use cases: rlog is most useful for exploratory analyses like PCA, heatmaps, and clustering, equal variance improves interpretability.
- Not for differential expression: It is not recommended for statistical testing (e.g., DESeq2 uses raw counts for DE analysis); rlog is mainly for visualization and QC.
- Computational note: rlog can be slower and memory-intensive for large datasets, so for big RNA-seq studies, variance stabilizing transformation (VST) is often preferred as a faster alternative.

Rscript PCA_PCC_using_rlog_data.R

```
(base) ibab@IBAB-MScBDB2-Comp007:~/NGS/23aug/5_HiSat2/4_feature_count$ Rscript PCA_PCC_using_rlog_data.R
Loading required package: S4Vectors
Loading required package: stats4
Loading required package: BiocGenerics

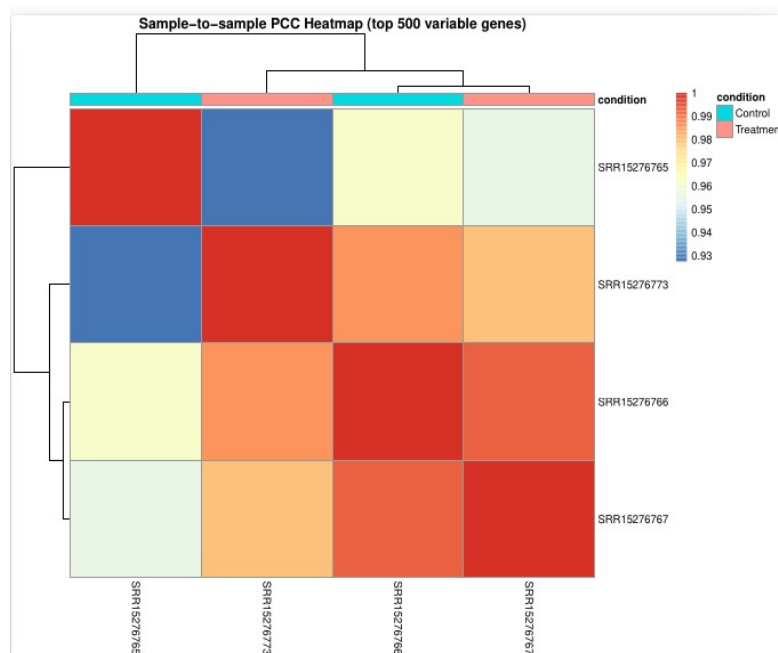
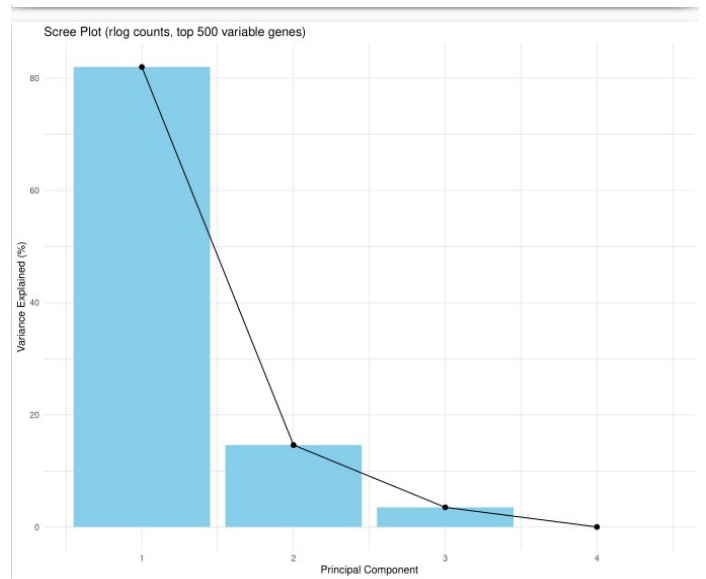
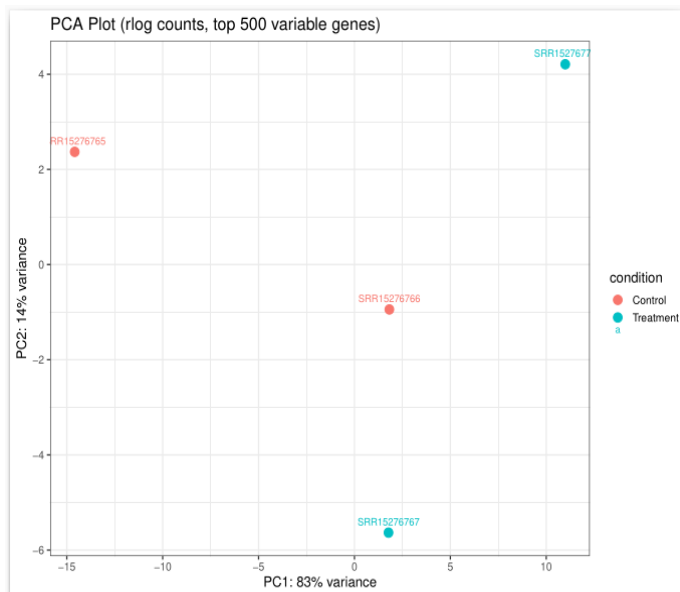
Attaching package: 'BiocGenerics'

The following objects are masked from 'package:stats':

  IQR, mad, sd, var, xtabs

The following objects are masked from 'package:base':
```

Output file: PCA_PCC_deseq2.pdf



Interpretation:

In the PCA plot, PC1 captures about 83 % of the total variance and cleanly separates the control samples (SRR15276765, SRR15276766) from the treatment samples (SRR15276767, SRR15276773), showing that the main source of variation is the experimental condition. The PC2, explaining about 14 % of the variance, reflects smaller differences within each group. The scree plot supports this, with PC1 explaining roughly 82–83 % of the variance, PC2 about 14 %, and later components each contributing less than 5 %. The sample-to-sample correlation heatmap shows very strong within-group correlations around 0.99 and slightly lower correlations between groups (~0.93–0.95).

Exercise 3: Read Count with STAR Alignments

1. Perform read count analysis on all 6 STAR-aligned BAM files using htseq-count.

```
cd 6_STAR
```

```
mkdir 3_htseq
```

```
mkdir 4_feature_count
```

```
cd 3_htseq
```

```
htseq-count SRR15276765Aligned.sortedByCoord.out.bam Pfalciparum.genome.gtf --idattr  
gene_id -t exon -f bam > SRR15276765_htseq.txt
```

Repeat the above command for all the other remaining samples also.

```
(base) ibab@IBAB-MScBD82-Comp007:~/NGS/23aug/6_STAR/3_htseq$ htseq-count /home/ibab/NGS/23aug/6_STAR/2_Alignment/SRR15276765Aligned.sortedByCoord.out.  
bam /home/ibab/NGS/23aug/2_Reference/Pfalciparum.genome.gtf --idattr gene_id -t exon -f bam > SRR15276765_htseq.txt  
35240 GFF lines processed.  
Warning: Read SRR15276765.1115947 claims to have an aligned mate which could not be found in an adjacent line.  
100000 alignment record pairs processed.  
200000 alignment record pairs processed.  
300000 alignment record pairs processed.  
400000 alignment record pairs processed.  
500000 alignment record pairs processed.  
600000 alignment record pairs processed.
```

```
PF3D7_0100100 8  
PF3D7_0100200 9  
PF3D7_0100300 0  
PF3D7_0100400 1  
PF3D7_0100500 3  
PF3D7_0100600 2  
PF3D7_0100700 0
```

Column-1: Gene ID

Column-2: Raw read count

SRR15276765_htseq.txt

```
__no_feature      3662999  
__ambiguous       3147  
__too_low_aQual   0  
__not_aligned     0  
__alignment_not_unique 311461
```

SRR15276766_htseq.txt

```
__no_feature      16724118  
__ambiguous       22571  
__too_low_aQual   0  
__not_aligned     0  
__alignment_not_unique 572133
```

SRR15276767_htseq.txt

```
__no_feature      19400296  
__ambiguous       27483  
__too_low_aQual   0  
__not_aligned     0  
__alignment_not_unique 446881
```

SRR15276771_htseq.txt

```
__no_feature      5282056  
__ambiguous       4254  
__too_low_aQual   0  
__not_aligned     0  
__alignment_not_unique 255001
```

SRR15276773_htseq.txt

```
__no_feature      19722189  
__ambiguous       23622  
__too_low_aQual   0  
__not_aligned     0  
__alignment_not_unique 364989
```

SRR15276774_htseq.txt

```
__no_feature      2752498  
__ambiguous       1274  
__too_low_aQual   0  
__not_aligned     0  
__alignment_not_unique 93766
```

2. Perform read count analysis on all 6 STAR-aligned BAM files using FeatureCount.

```
cd ..
```

```
cd 4_feature_count
```

```
featureCounts -p -t exon -g gene_id -a Pfalciparum.genome.gtf  
SRR15276765Aligned.sortedByCoord.out.bam -o SRR15276765.txt
```

Repeat the above command for all the other remaining samples also.

```
(base) ibab@IBAB-MScBDB2-Comp007:~/NGS/23aug/6_STAR/4_feature_count$ featureCounts -p -t exon -g gene_id -a /home/ibab/NGS/23aug/2_Reference/Pfalciparum.genome.gtf /home/ibab/NGS/23aug/6_STAR/2_Alignment/SRR15276765Aligned.sortedByCoord.out.bam -o SRR15276765.txt
```



```
//===== FeatureCounts setting =====\\  
||  
||      Input files : 1 BAM file  
||  
||      SRR15276765Aligned.sortedByCoord.out.bam  
||  
||      Output file : SRR15276765.txt  
||      Summary : SRR15276765.txt.summary  
||      Paired-end : yes  
||      Count read pairs : no  
||      Annotation : Pfalciparum.genome.gtf (GTF)  
||      Dir for temp files : ./  
||
```

```
(base) ibab@IBAB-MScBDB2-Comp007:~/NGS/23aug/6_STAR/4_feature_count$ ls -lh  
total 3.3M  
-rw-rw-r-- 1 ibab ibab 544K Sep 20 13:22 SRR15276765.txt  
-rw-rw-r-- 1 ibab ibab 437 Sep 20 13:22 SRR15276765.txt.summary  
-rw-rw-r-- 1 ibab ibab 547K Sep 20 13:24 SRR15276766.txt  
-rw-rw-r-- 1 ibab ibab 440 Sep 20 13:24 SRR15276766.txt.summary  
-rw-rw-r-- 1 ibab ibab 547K Sep 20 13:25 SRR15276767.txt  
-rw-rw-r-- 1 ibab ibab 441 Sep 20 13:25 SRR15276767.txt.summary  
-rw-rw-r-- 1 ibab ibab 545K Sep 20 13:25 SRR15276771.txt  
-rw-rw-r-- 1 ibab ibab 437 Sep 20 13:25 SRR15276771.txt.summary  
-rw-rw-r-- 1 ibab ibab 547K Sep 20 13:25 SRR15276773.txt  
-rw-rw-r-- 1 ibab ibab 440 Sep 20 13:25 SRR15276773.txt.summary  
-rw-rw-r-- 1 ibab ibab 543K Sep 20 13:26 SRR15276774.txt  
-rw-rw-r-- 1 ibab ibab 437 Sep 20 13:26 SRR15276774.txt.summary
```

SRR15276765.txt.summary

Assigned	3795427
Unassigned_Unmapped	0
Unassigned_Read_Type	0
Unassigned_Singleton	0
Unassigned_MappingQuality	0
Unassigned_Chimera	0
Unassigned_FragmentLength	0
Unassigned_Duplicate	0
Unassigned_MultiMapping	730121
Unassigned_Secondary	0
Unassigned_NonSplit	0
Unassigned_NoFeatures	139885
Unassigned_Overlapping_Length	0
Unassigned_Ambiguity	49614

SRR15276766.txt.summary

Assigned	17226580
Unassigned_Unmapped	0
Unassigned_Read_Type	0
Unassigned_Singleton	0
Unassigned_MappingQuality	0
Unassigned_Chimera	0
Unassigned_FragmentLength	0
Unassigned_Duplicate	0
Unassigned_MultiMapping	1433457
Unassigned_Secondary	0
Unassigned_NonSplit	0
Unassigned_NoFeatures	892621
Unassigned_Overlapping_Length	0
Unassigned_Ambiguity	245986

SRR15276767.txt.summary

Assigned	19987650
Unassigned_Unmapped	0
Unassigned_Read_Type	0
Unassigned_Singleton	0
Unassigned_MappingQuality	0
Unassigned_Chimera	0
Unassigned_FragmentLength	0
Unassigned_Duplicate	0
Unassigned_MultiMapping	1189432
Unassigned_Secondary	0
Unassigned_NonSplit	0
Unassigned_NoFeatures	1052521
Unassigned_Overlapping_Length	0
Unassigned_Ambiguity	290399

SRR15276771.txt.summary

```
Assigned      5388528
Unassigned_Unmapped      0
Unassigned_Read_Type      0
Unassigned_Singleton      0
Unassigned_MappingQuality      0
Unassigned_Chimera      0
Unassigned_FragmentLength      0
Unassigned_Duplicate      0
Unassigned_MultiMapping 642874
Unassigned_Secondary      0
Unassigned_NonSplit      0
Unassigned_NoFeatures 197796
Unassigned_Overlapping_Length      0
Unassigned_Ambiguity      79551
```

SRR15276773.txt.summary

```
Assigned      20003361
Unassigned_Unmapped      0
Unassigned_Read_Type      0
Unassigned_Singleton      0
Unassigned_MappingQuality      0
Unassigned_Chimera      0
Unassigned_FragmentLength      0
Unassigned_Duplicate      0
Unassigned_MultiMapping 1006175
Unassigned_Secondary      0
Unassigned_NonSplit      0
Unassigned_NoFeatures 932947
Unassigned_Overlapping_Length      0
Unassigned_Ambiguity      306433
```

SRR15276774.txt.summary

```
Assigned      2689141
Unassigned_Unmapped      0
Unassigned_Read_Type      0
Unassigned_Singleton      0
Unassigned_MappingQuality      0
Unassigned_Chimera      0
Unassigned_FragmentLength      0
Unassigned_Duplicate      0
Unassigned_MultiMapping 250982
Unassigned_Secondary      0
Unassigned_NonSplit      0
Unassigned_NoFeatures 103307
Unassigned_Overlapping_Length      0
Unassigned_Ambiguity      40634
```

3. For htseq-count data:

- Generate Boxplot, PCA, and PCC plots with and without CPM filtering.

```
(
    echo -e "Geneid\tSRR15276765\tSRR15276766\tSRR15276767\tSRR15276771\t
    tSRR15276773\tSRR15276774"

    paste *_htseq.txt \

    | awk 'BEGIN{OFS="\t"} $1 ~ /^PF3D7/ {printf $1; for(i=2;i<=NF;i+=2) printf "\t"$i; print ""}'

) > raw_read.txt
```

This command merges all *_htseq.txt files into a single tab-delimited matrix containing only rows whose gene IDs start with PF3D7. It first prints a header with sample names, then extracts the gene ID and count columns from each file, and saves everything to raw_read.txt.

```
(base) ibab@IBAB-MScBDB2-Comp007:~/NGS/23aug/6_STAR/3_htseq$ wc -l raw_read.txt
5681 raw_read.txt
(base) ibab@IBAB-MScBDB2-Comp007:~/NGS/23aug/6_STAR/3_htseq$ head -4 raw_read.txt
Geneid SRR15276765 SRR15276766 SRR15276767 SRR15276771 SRR15276773 SRR15276774
PF3D7_0100100 8 95 110 8 76 12
PF3D7_0100200 9 24 10 10 15 0
PF3D7_0100300 0 0 4 0 0 0
```

Rscript BOX_PCA_PCC_without_CPM_filtering.R

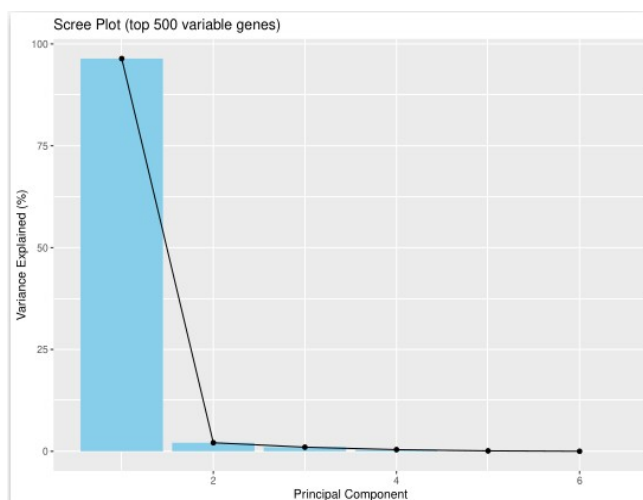
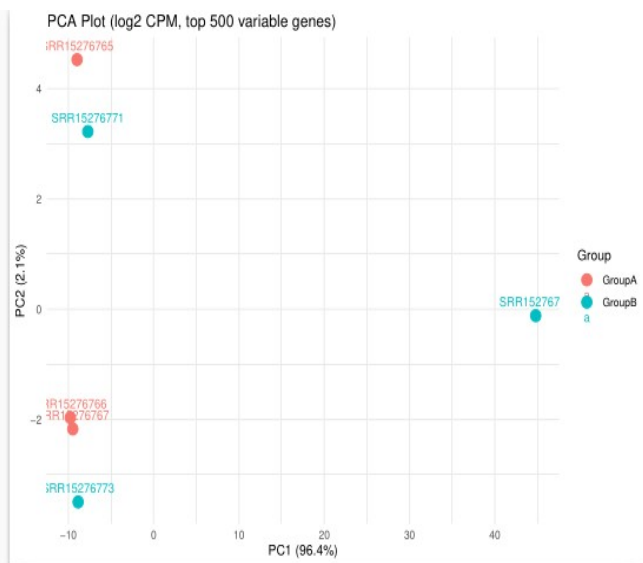
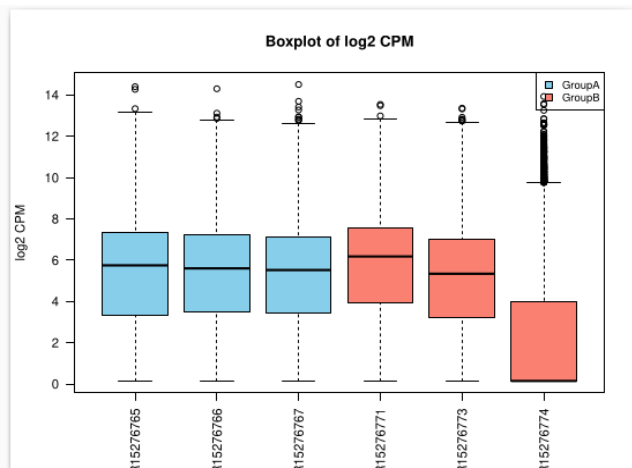
```
(base) ibab@IBAB-MScBDB2-Comp007:~/NGS/23aug/6_STAR/3_htseq$ Rscript BOX_PCA_PCC_without_CPM_filtering.R
Loading required package: limma
null device
1
✓ All plots saved in QC_plots.pdf
```

Rscript BOX_PCC_PCA_with_CPM_filtering.R

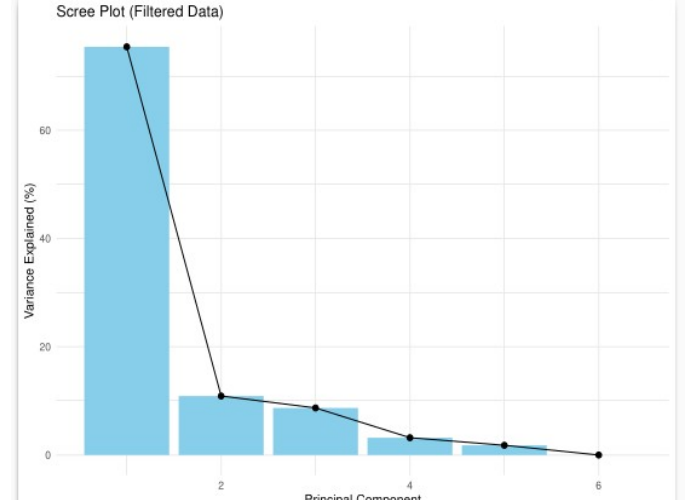
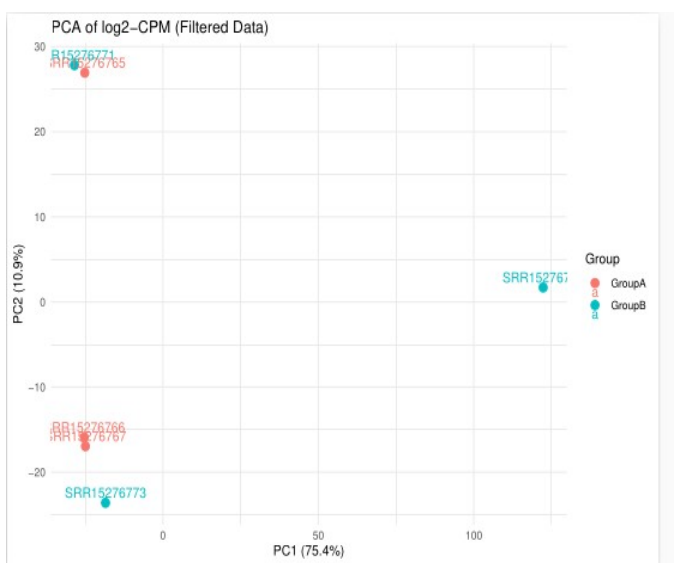
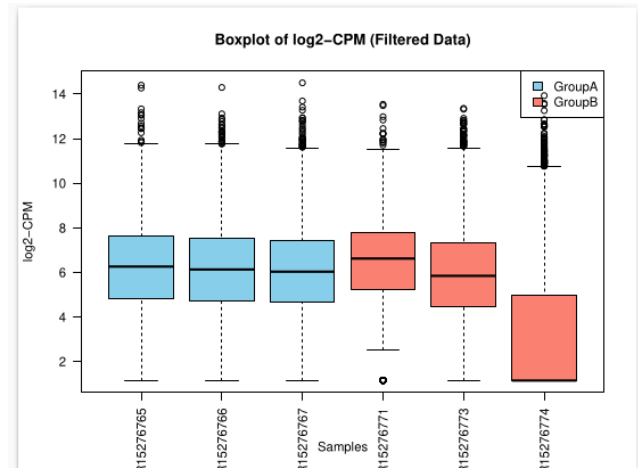
```
# Step 2: CPM filtering
#-----
cpm_counts <- cpm(dge)
keep <- rowSums(cpm_counts > 1) >= 4
dge_filtered <- dge[keep, , keep.lib.sizes=FALSE]
```

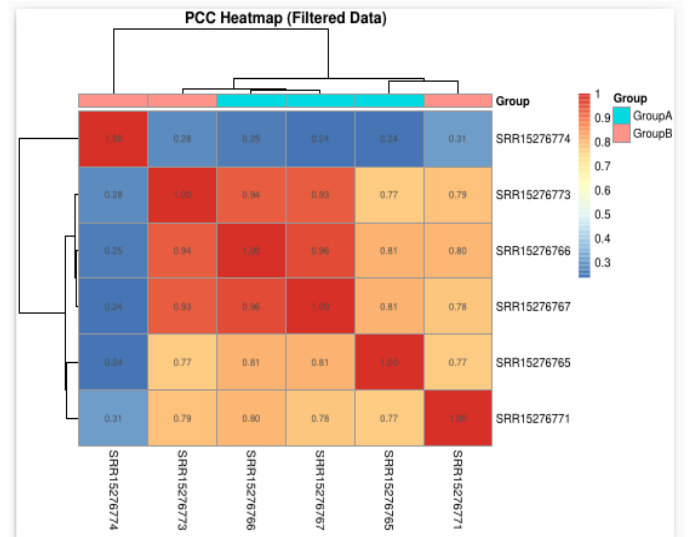
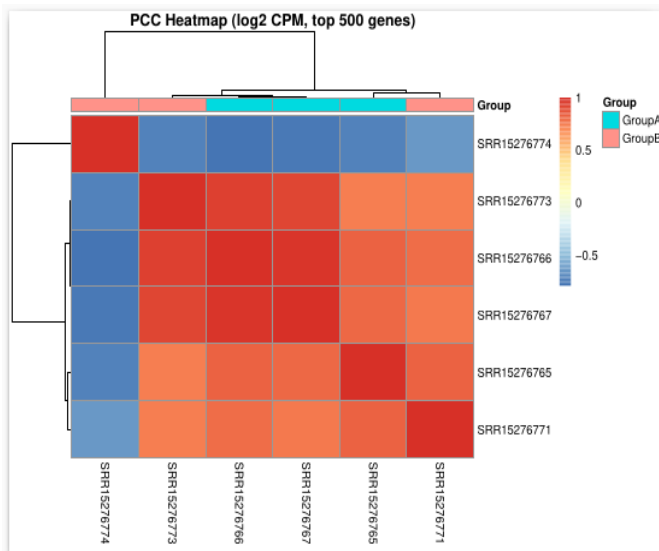
```
(base) ibab@IBAB-MScBDB2-Comp007:~/NGS/23aug/6_STAR/3_htseq$ Rscript BOX_PCC_PCA_with_CPM_filtering.R
Loading required package: limma
Genes before filtering: 5680
Genes after filtering: 4775
null device
1
```

Without CPM filtering



With CPM filtering

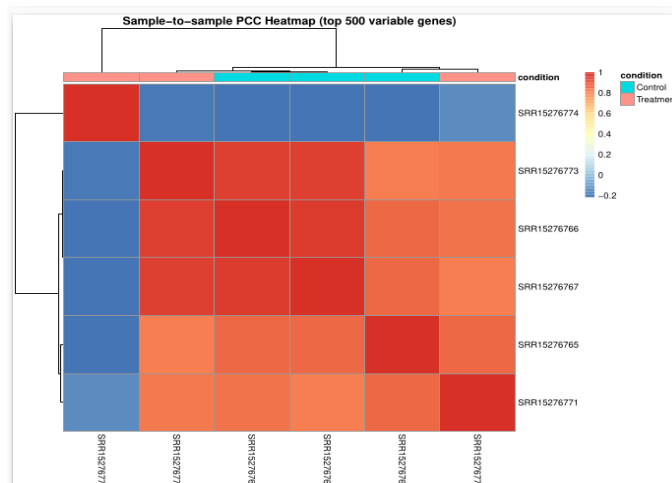
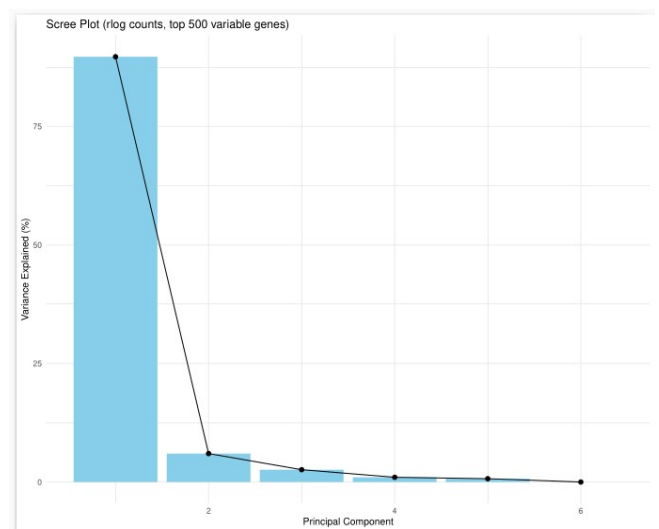
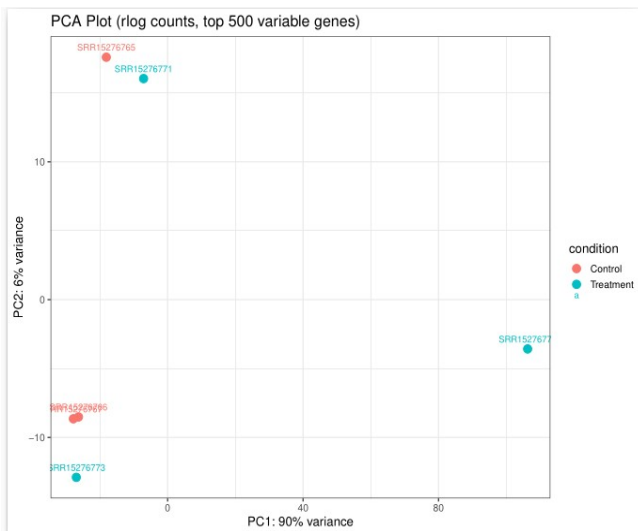




➤ Generate PCA and PCC plots using rlog filtering.

Rscript PCA_PCC_using_rlog_data.R

```
(base) lib64R/lib64R-M5c8082-Comp007:~/NGS/23aug/6_STAR/3_htseq$ Rscript PCA_PCC_using_rlog_data.R
Loading required package: S4Vectors
Loading required package: stats4
Loading required package: BiocGenerics
Attaching package: 'BiocGenerics'
The following objects are masked from 'package:stats':
```



4. For FeatureCount data:

- **Generate Boxplot, PCA, and PCC plots with and without CPM filtering.**

```
cd ..
```

```
cd 4_feature_count
```

```
(base) ibab@IBAB-MScBDB2-Comp007:~/NGS/23aug/6_STAR/4_feature_count$ cat SRR15276765.txt
# Program:featureCounts v2.0.6; Command:"featureCounts" "-p" "-t" "exon" "-g" "gene_id" "-a" "/home/ibab/NGS/23aug/2_Reference/Pfalciparum.genome.gtf"
"/home/ibab/NGS/23aug/6_STAR/2_Alignment/SRR15276765Aligned.sortedByCoord.out.bam" "-o" "SRR15276765.txt"
Geneid Chr Start End Strand Length /home/ibab/NGS/23aug/6_STAR/2_Alignment/SRR15276765Aligned.sortedByCoord.out.bam
malmito_rna_16:rRNA M76611 3 33 + 31 0
malmito_rna_20:rRNA M76611 34 71 + 38 0
malmito_rna_9:rRNA M76611 72 125 + 54 0
malmito_rna_17:rRNA M76611 126 165 + 40 0
```

```
awk '

```

```
/^#/ {next}

```

```
header==0 {

```

```
sample="SRR15276765"

```

```
print "Geneid\t"sample

```

```
header=1

```

```
next

```

```
}

```

```
/^PF3D7/ {print $1"\t"$NF}

```

```
' SRR15276765.txt > PF3D7_sample1.txt
```

Repeat this command for other remaining samples also.

```
Geneid SRR15276765
PF3D7_0100100 145
PF3D7_0100200 56
PF3D7_0100300 4
PF3D7_0100400 12
```

```
paste PF3D7_*.txt > merged.txt
```

```
Geneid SRR15276765 Geneid SRR15276766 Geneid SRR15276767 Geneid SRR15276771 Geneid SRR15276773 Geneid SRR15276774
PF3D7_0100100 145 PF3D7_0100100 956 PF3D7_0100100 945 PF3D7_0100100 207 PF3D7_0100100 783 PF3D7_0100100 97
PF3D7_0100200 56 PF3D7_0100200 218 PF3D7_0100200 75 PF3D7_0100200 59 PF3D7_0100200 122 PF3D7_0100200 44
PF3D7_0100300 4 PF3D7_0100300 6 PF3D7_0100300 6 PF3D7_0100300 4 PF3D7_0100300 2 PF3D7_0100300 0
PF3D7_0100400 12 PF3D7_0100400 65 PF3D7_0100400 107 PF3D7_0100400 30 PF3D7_0100400 124 PF3D7_0100400 74
```

```
cut -f1,2,4,6,8,10,12 merged.txt > raw_read.txt
```

```
(base) ibab@IBAB-MScBDB2-Comp007:~/NGS/23aug/6_STAR/4_feature_count$ wc -l raw_read.txt
5681 raw_read.txt
(base) ibab@IBAB-MScBDB2-Comp007:~/NGS/23aug/6_STAR/4_feature_count$ head -4 raw_read.txt
Geneid SRR15276765 SRR15276766 SRR15276767 SRR15276771 SRR15276773 SRR15276774
PF3D7_0100100 145 956 945 207 783 97
PF3D7_0100200 56 218 75 59 122 44
PF3D7_0100300 4 6 6 4 2 0
```

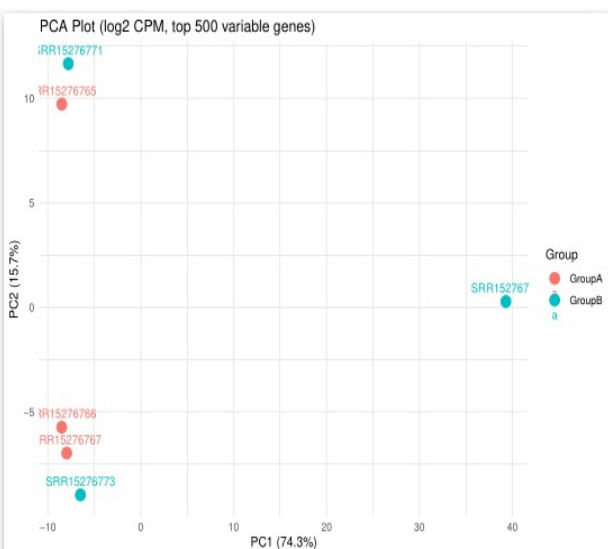
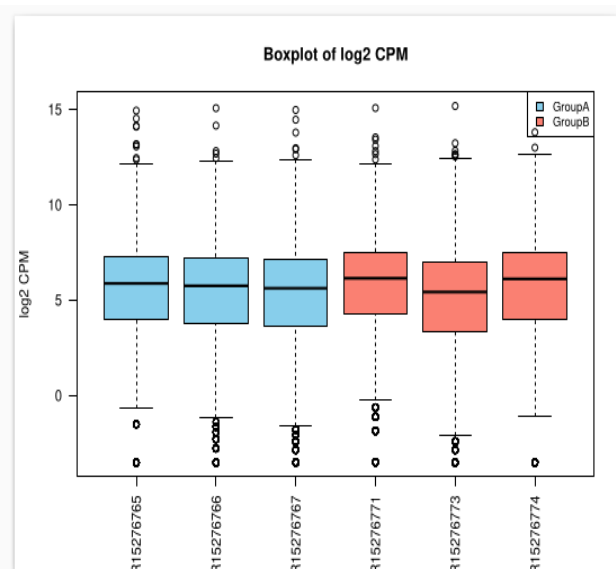

Rscript BOX_PCA_PCC_without_CPM_filtering.R

```
(base) ibab@IBAB-MScBDB2-Comp007:~/NGS/23aug/6_STAR/4_feature_count$ Rscript BOX_PCA_PCC_without_CPM_filtering.R
Loading required package: limma
null device
1
✓ All plots saved in QC_plots.pdf
```

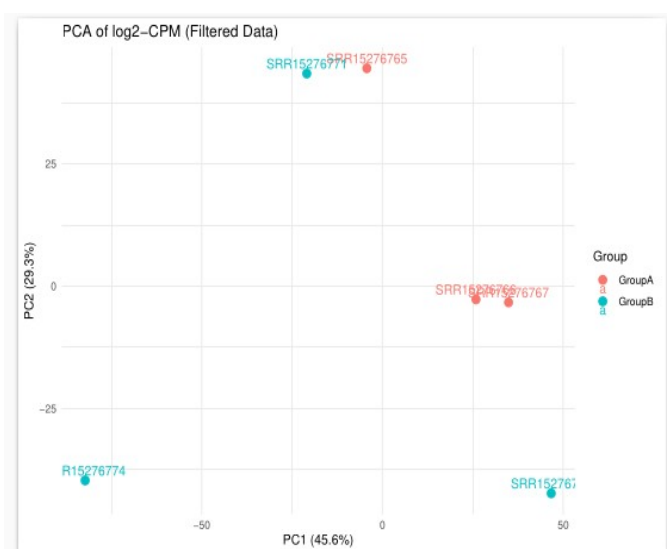
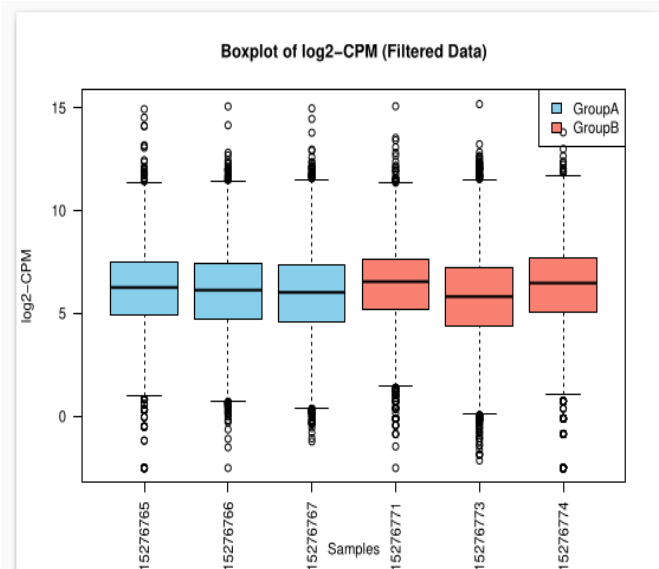
Rscript BOX_PCC_PCA_with_CPM_filtering.R

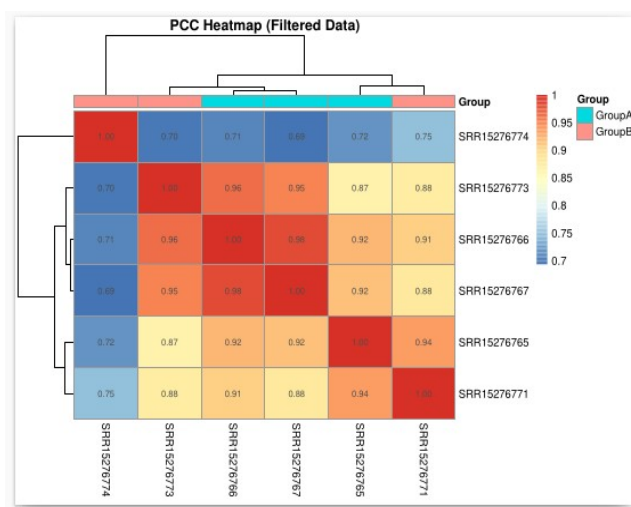
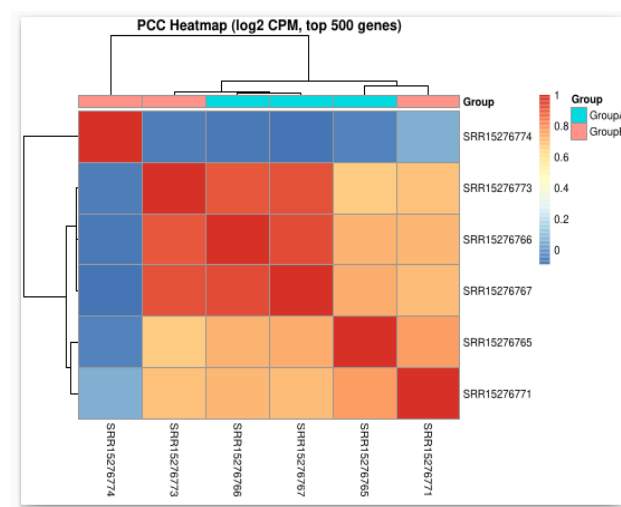
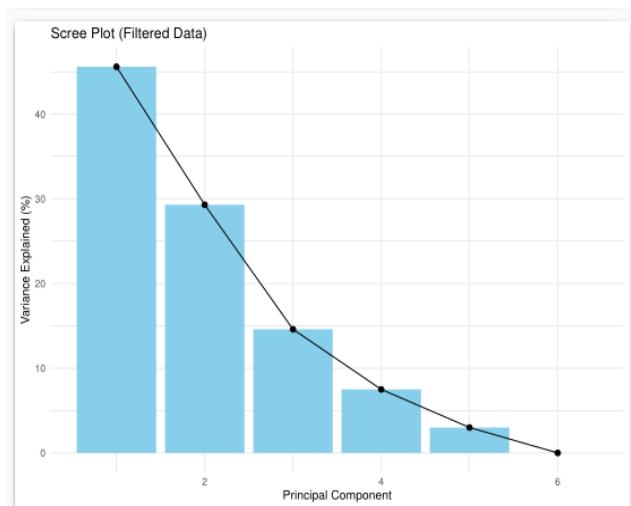
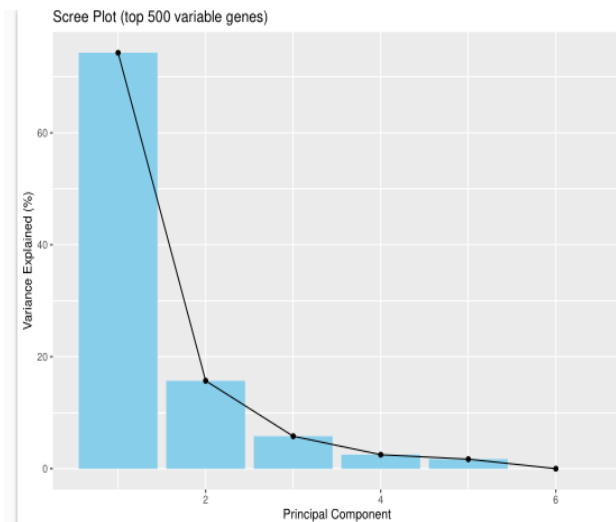
```
(base) ibab@IBAB-MScBDB2-Comp007:~/NGS/23aug/6_STAR/4_feature_count$ Rscript BOX_PCC_PCA_with_CPM_filtering.R
Loading required package: limma
Genes before filtering: 5680
Genes after filtering: 4951
null device
1
```

Without CPM filtering



With CPM filtering





➤ Generate PCA and PCC plots using rlog filtering.

Rscript PCA_PCC_using_rlog_data.R

```
(base) ibab@IBAB-MScBDB2-Comp007: ~/NGS/23aug/6_STAR/4_feature_count$ Rscript PCA_PCC_using_rlog_data.R
Loading required package: S4Vectors
Loading required package: stats4
Loading required package: BiocGenerics
Attaching package: 'BiocGenerics'
```

